

Orion Relay Design

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1. Introduction

Product `model` for resilient remote store-and-forward communications.

1.1. Scope and Assumptions

- `Functional Groups` and `Functional Units` are the stable authored design backbone.
- `Runtime elements` and `code elements` are inferred from available IaC, deployment, and source artifacts.
- Inferred coverage is evidence-driven and may be partial when source metadata is missing.
- `Requirements` and reference indexes are provided for traceability, not as primary narrative flow.

2. How To Read This Document

This document is intentionally split into two layers:

- 1. Concept layer: functional and view-specific architecture narratives for human understanding.
- 2. Evidence layer: generated requirement traceability and inferred reference indexes.

Read in this order:

- 1. Architecture Intent View: stable authored architecture intent and ownership boundaries.
- 2. Communication / Deployment / Security Views: viewpoint-specific operational behavior.
- 3. Traceability View: requirement mapping, evidence coverage, and confidence.
- 4. Reference Appendix: supporting reference IDs and inferred-source links.

3. Document Health Snapshot

View	Authored Status	Authored Status Explanation	Units in Scope	Direct Evidence Coverage	Gap Count	Heuristic Confidence	Heuristic Basis
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4. Terms and Definitions

Term	Definition
actor (TERM-ACTOR)	A person or operational role that interacts with functional units.
architecture decision (TERM-DECISION)	Authored architecture decision record with status, context, decision text, and consequences.
AsciiDoc diagram (TERM-ASCIIDOC-DIAGRAM)	Generated diagram block, usually Mermaid, that visualizes architecture relationships in the AsciiDoc publication.
AsciiDoc document (TERM-ASCIIDOC-DOCUMENT)	Human-readable generated architecture publication document assembled from authored model, inferred evidence, diagrams, references, and decisions.
AsciiDoc section (TERM-ASCIIDOC-SECTION)	Structured generated document section or row model used to render authored and inferred architecture content.
attack vector (TERM-ATTACK-VECTOR)	A technical misuse or attack path that targets functional, referenced, or runtime elements.
authored mapping (TERM-AUTHORED-MAPPING)	An explicit relationship declared in architecture inputs between authored or referenced elements.
catalog (TERM-CATALOG)	Controlled vocabulary document used to normalize model, requirement, and generated-document terminology.
catalog entry (TERM-CATALOG-ENTRY)	A stable catalog term with a definition and aliases for linting, linking, and generated references.

Term	Definition
CLI command (TERM-CLI-COMMAND)	Command-line entrypoint that coordinates loading, validation, generation, and file output for an engineering-model workflow.
code element (TERM-CODE-ELEMENT)	An inferred code structure or ownership element discovered from source trees and build metadata.
compliance mapping (TERM-COMPLIANCE-MAPPING)	Authored mapping that links a model control to selected compliance controls, model scope, implementation status, evidence, and responsible roles.
control (TERM-CONTROL)	An authored security, policy, or operational control used to constrain behavior and reduce risk.
control verification (TERM-CONTROL-VERIFICATION)	Authored control verification record with method, status, threat/risk scope, findings, and evidence.
data object (TERM-DATA-OBJECT)	An authored data artifact or contract shape traced across flow, interface, and implementation boundaries.
deployment target (TERM-DEPLOYMENT-TARGET)	An authored deployment destination such as environment, cluster, namespace, or registry scope.
design document (TERM-DESIGN)	Authored design narrative organized by model entities and architecture views.
design view (TERM-DESIGN-VIEW)	View-scoped design narrative attached to an authored functional group or functional unit.
event (TERM-EVENT)	An authored trigger signal that drives transitions, flow progress, or asynchronous outcomes.
evidence (TERM-EVIDENCE)	A file, artifact, or observation used to support control, risk, threat, or verification claims.
flow (TERM-FLOW)	An authored causal interaction sequence from user/system intent to outcome, represented as ordered steps.
flow step (TERM-FLOW-STEP)	A single authored step in a flow that captures action, data in/out, references, and normal/error/async transitions.
functional group (TERM-FUNCTIONAL-GROUP)	A major authored capability area that groups related functional units.
functional unit (TERM-FUNCTIONAL-UNIT)	An authored working unit inside a functional group that owns specific behavior.
generation workflow (TERM-GENERATION-WORKFLOW)	Orchestrated run that combines model loading, validation, projection, rendering, and artifact emission.

Term	Definition
inference hint (TERM-INFERENCE-HINT)	Authored source and ownership configuration used to discover runtime, code, and verification evidence.
inferred mapping (TERM-INFERRED-MAPPING)	A discovered relationship that links inferred runtime/code elements upward to authored design.
interface (TERM-INTERFACE)	An authored interface boundary (for example API, channel, endpoint, or contract) owned by a functional unit.
link scheduler software (FU-ORI-LINK)	Selects available radio links and transmission windows.
lint run (TERM-LINT-RUN)	Requirement lint configuration that controls parsing and quality checks for requirement text.
LOBSTER activity trace (TERM-LOBSTER-ACTIVITY-TRACE)	Generated LOBSTER activity JSON that links verification evidence back to formal requirement identifiers.
MCP server (TERM-MCP-SERVER)	Model Context Protocol server that exposes engineering-model operations to AI agents through JSON-RPC tools.
MCP stdio transport (TERM-MCP-STDIO-TRANSPORT)	Content-Length framed standard input/output transport used by the MCP command-line server.
MCP tool (TERM-MCP-TOOL)	Named MCP operation with a constrained input schema and structured response payload.
MCP tool response (TERM-MCP-TOOL-RESPONSE)	Structured MCP tool payload that includes success state, schema version, generated timestamp, and result data or validation error.
message flooding (AV-ORI-MESSAGE-FLOOD)	Excess messages attempt to exhaust constrained link capacity.
message router software (FU-ORI-ROUTER)	Stores, prioritizes, and forwards operational messages.
model (TERM-MODEL)	The authored architecture model root that composes functional structure, relationships, inference hints, and views.
model bundle (TERM-BUNDLE)	Loaded architecture, catalog, and companion documents resolved as one working model context.
network controller (ACT-ORI-CONTROLLER)	Configures relay priorities and monitors delivery.
Open OTM document (TERM-OPEN-OTM-DOCUMENT)	Open Threat Model JSON representation generated from the engineering model for interoperable threat-model exchange.

Term	Definition
orion relay system (SYS-ORI)	Resilient remote communications and store-and-forward product.
OSCAL assessment results (TERM-OSCAL-ASSESSMENT-RESULTS)	Generated OSCAL assessment results that summarize reviewed controls, verification findings, and modeled risks.
OSCAL POA&M (TERM-OSCAL-POAM)	Generated OSCAL plan of action and milestones that maps modeled POA&M items and related risks into compliance JSON.
OSCAL SSP (TERM-OSCAL-SSP)	Generated OSCAL system security plan that maps authored system metadata, components, controls, allocations, and evidence into SSP JSON.
POA&M item (TERM-POAM-ITEM)	Plan of action and milestones item linked to a modeled risk and supporting artifacts.
reference index (TERM-REFERENCE-INDEX)	Generated index of authored, catalog, runtime, code, and verification references with stable anchors and backlinks.
referenced element (TERM-REFERENCED-ELEMENT)	An architecture-relevant external, platform, or third-party dependency represented by role.
relay message (DATA-ORI-MESSAGE)	Prioritized payload awaiting delivery across a remote link.
relay mission (FG-ORI)	Product-specific link scheduling and message delivery.
relay security engineer (ACT-ORI-ENGINEER)	Owns Orion security controls and risk.
repository index (TERM-REPO-INDEX)	Bounded first-party source index used to answer model-aware file, requirement, control, and threat lookup queries.
requirement (TERM-REQUIREMENT)	A structured requirement statement that defines expected system behavior and maps to functional ownership.
risk (TERM-RISK)	Authored risk record with likelihood, impact, response, scope, controls, and evidence.
runtime element (TERM-RUNTIME-ELEMENT)	An inferred runtime realization element discovered from infrastructure and deployment sources.
security context (TERM-SECURITY-CONTEXT)	Security-focused generated context view that groups owned functional units and shows external actors, references, flows, controls, and trust boundaries.
source block (TERM-SOURCE-BLOCK)	Stable source reference block that records file path, optional line span, kind, summary, and linked entity IDs.
state (TERM-STATE)	An authored lifecycle state used to model transition behavior, guards, and event-driven progression.

Term	Definition
Structurizr DSL (TERM-STRUCTURIZR-DSL)	Generated Structurizr DSL workspace that projects authored architecture, relationships, dynamic views, and deployment metadata.
threat assumption (TERM-THREAT-ASSUMPTION)	Authored security assumption that records scope, rationale, owner, status, and evidence.
Threat Dragon document (TERM-THREAT-DRAGON-DOCUMENT)	Threat Dragon JSON representation generated from the engineering model for STRIDE threat-model review.
threat mitigation (TERM-THREAT-MITIGATION)	Authored mitigation record that links a threat scenario to a control, effectiveness, verification, and evidence.
threat model export (TERM-THREAT-MODEL-EXPORT)	Generated security artifact that translates authored architecture, flows, trust boundaries, threats, and mitigations into an external threat-model schema.
threat out-of-scope decision (TERM-THREAT-OUT-OF-SCOPE)	Authored decision that excludes a threat concern from current scope with reason, owner, expiry, and evidence.
threat scenario (TERM-THREAT-SCENARIO)	Authored threat narrative that connects attack vectors, scope, flows, controls, risks, and verification evidence.
trace marker (TERM-TRACE-MARKER)	Source-level marker such as TRLC-LINKS or ENGMODEL-LINKS used to connect declarations to requirements and model entities.
TRLC package (TERM-TRLC-PACKAGE)	Generated TRLC model and requirements package used for formal requirement traceability processing.
trust boundary (TERM-TRUST-BOUNDARY)	An authored trust separation zone that marks policy, identity, or security control boundaries.
upward linking (TERM-UPWARD-LINKING)	Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.
validation (TERM-VALIDATION)	Model quality gate that checks authored documents, references, relationship semantics, and requirement lint results.
validation diagnostic (TERM-VALIDATION-DIAGNOSTIC)	Structured validation finding with code, severity, message, and optional source path.
verification check (TERM-VERIFICATION-CHECK)	An inferred or authored verification artifact that validates requirement behavior with test evidence.
view (TERM-VIEW)	Authored projection configuration that selects roots, entity kinds, mappings, audience, and abstraction.

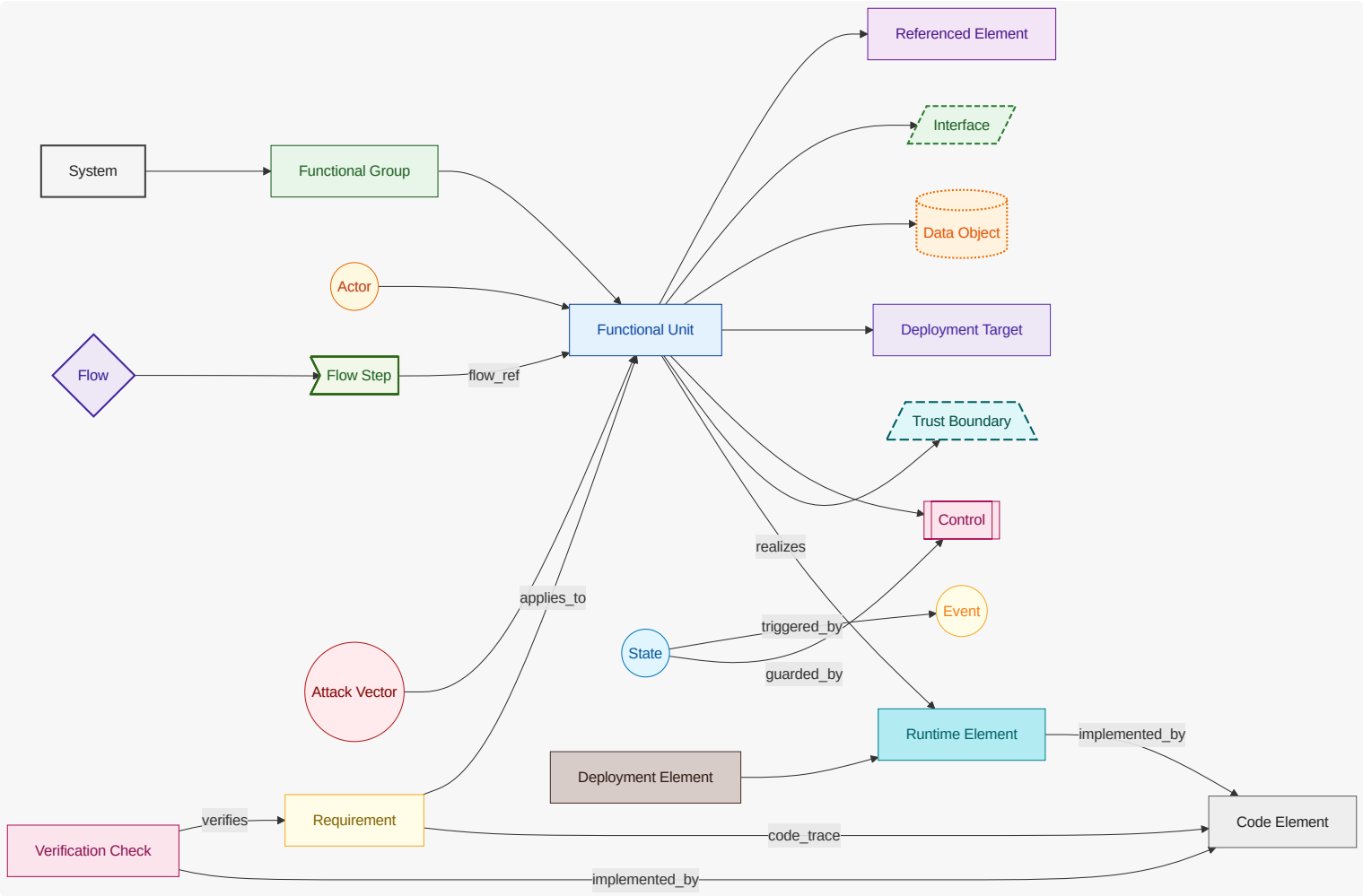
5. Architecture Decision Records

Architecture decision records are generated separately and linked here as a compact index.

ID	Title	Status	Date
<u>ADR-ORI-001</u>	Keep adaptive radio scheduling product-specific	accepted	2026-09-19

5.1. Diagram Color and Shape Legend

This legend applies to all Mermaid diagrams in this document.



6. Traceability Appendix

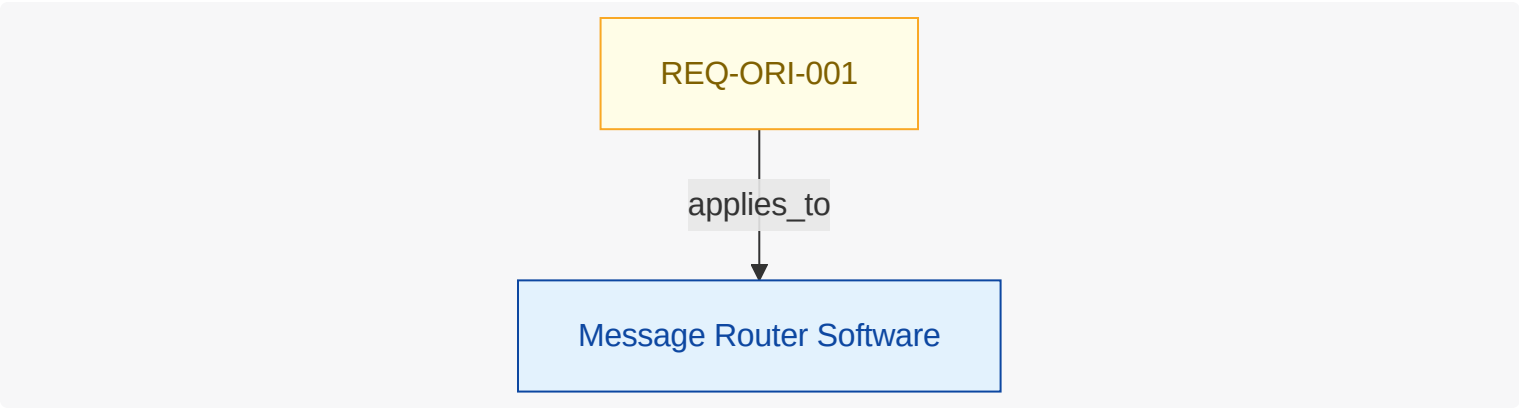
6.1. Requirement Details

6.1.1. REQ-ORI-001

The orion relay system shall deliver each relay message according to network controller priority.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.

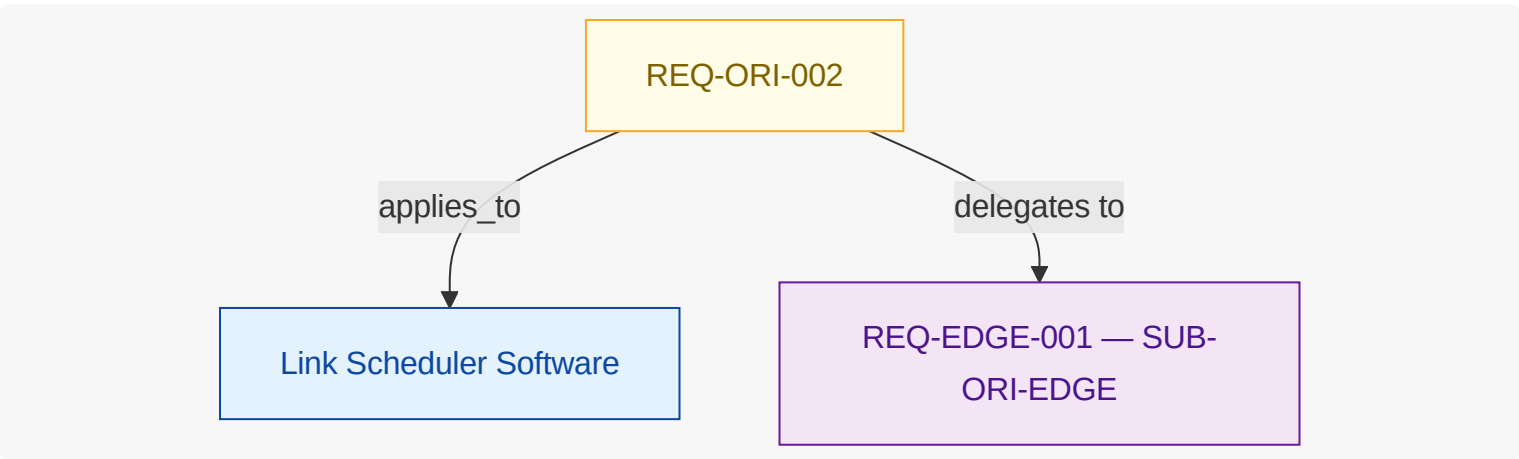


6.1.2. REQ-ORI-002

The orion relay system shall execute the link scheduler software on managed edge compute.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.

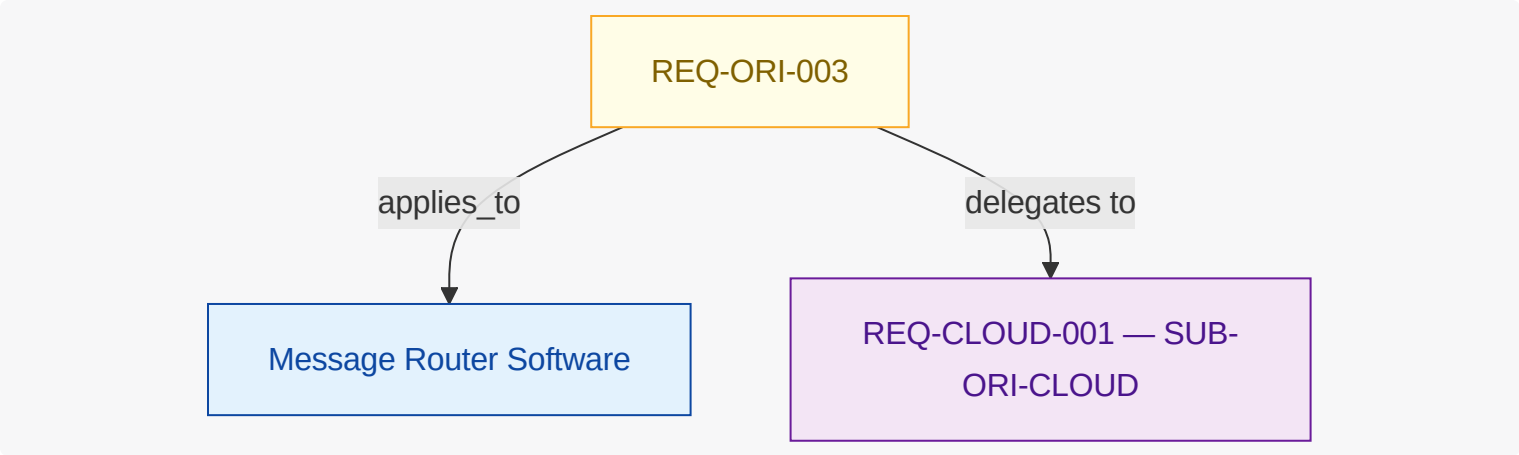


6.1.3. REQ-ORI-003

The orion relay system shall retain relay configuration using managed cloud operations.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.

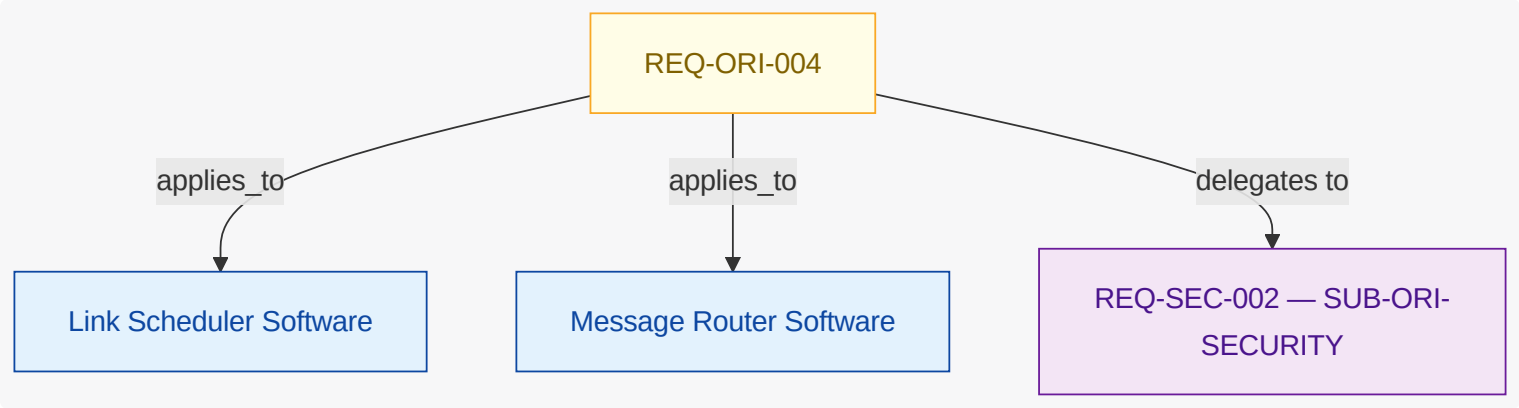


6.1.4. REQ-ORI-004

The orion relay system shall authenticate product services using shared security services.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.

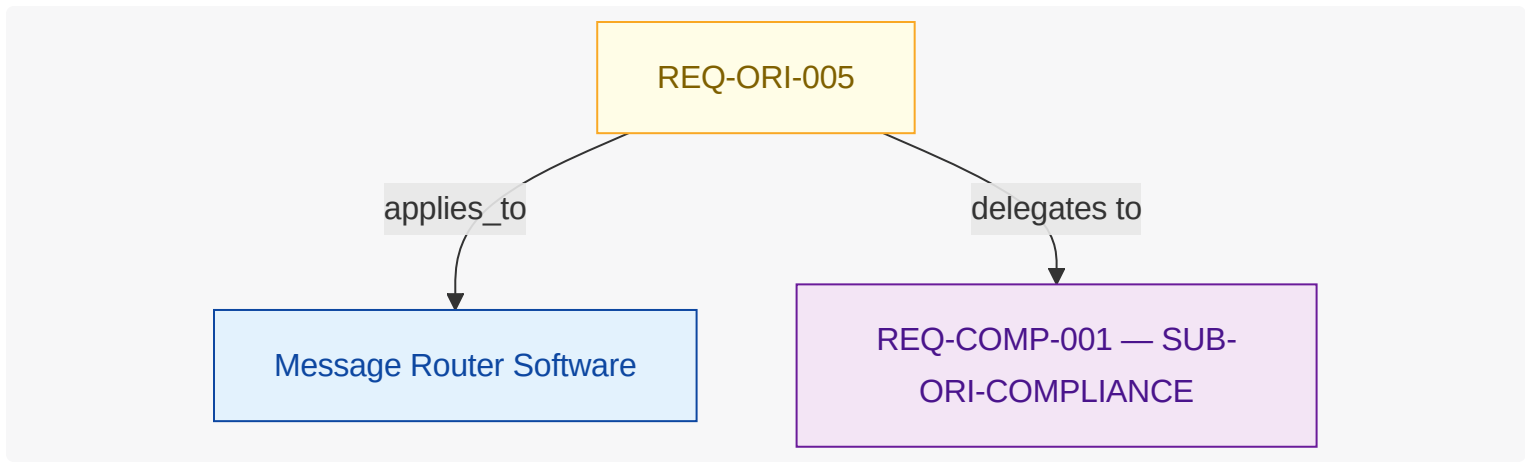


6.1.5. REQ-ORI-005

The orion relay system shall retain compliance evidence using the company control baseline.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



6.2. Verification Inventory

What this section shows: verification checks inferred from test artifacts, linked to test code elements and mapped to verified requirements.

7. Gemara GRC Model

This chapter renders the model as OpenSSF Gemara (<https://gemara.openssf.org>) documents. Each artifact is built with the official `go-gemara` SDK types and validated against the published Gemara CUE schemas. Generate the full YAML set with `enggemara` ; query it via the `gemara.*` MCP tools.

Gemara Layer	Artifact	Entries
L1	Vector Catalog	1
L1	Principle Catalog	5
L1	Guidance Catalog	1
L2	Capability Catalog	3
L2	Control Catalog	1
L2	Threat Catalog	0
L3	Risk Catalog	1
L3	Policy	1
-	Lexicon	8
L7	Audit Log	2
L6	Enforcement Log	1

7.1. Controls (L2)

Control	Objective	Group	Assessment Requirements
CTRL-ORI-ADMISSION	Reserves capacity for authenticated priority messages.	availability	1

7.2. Risks (L3)

Risk	Title	Severity
RISK-ORI-LINK-EXHAUSTION	Relay link exhaustion	High

8. System Composition & Hardware

This chapter shows the system architecture across hardware and software: hardware items and their interfaces, the subsystems this system composes, and the allocation of this system’s requirements onto them. Subsystem references are downward-only and resolved from exact-version manifest dependencies, with optional workspace replacements.

8.1. Hardware Items

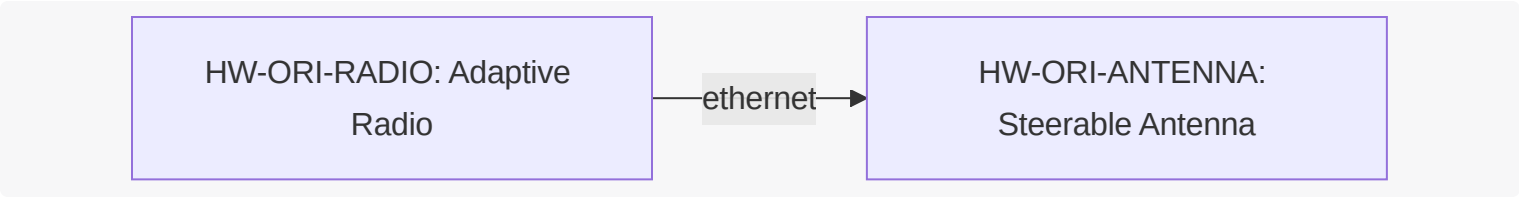
ID	Name	Kind	Hosts	Safety
HW-ORI-RADIO	Adaptive Radio	lru		-
HW-ORI-ANTENNA	Steerable Antenna	actuator		-

8.2. Hardware Interfaces

ID	Bus	From → To	Dir	Software IF
HWIF-ORI-RF-CONTROL	ethernet	HW-ORI-RADIO → HW-ORI-ANTENNA	bidirectional	IF-ORI-MESSAGE

8.2.1. Hardware/Software Interface View

Hardware IDs identify the items in the tables above. The Hardware Items table lists every hosted unit.



8.3. Subsystems

ID	Name	Reference
SUB-ORI-EDGE	Shared Edge Platform	edge::public
SUB-ORI-CLOUD	Shared Cloud Platform	cloud::public
SUB-ORI-SECURITY	Shared Security Services	security::public
SUB-ORI-COMPLIANCE	Shared Compliance Baseline	compliance::public

8.4. Selected Dependency Publications

Alias	Publication	Module	Version	Source model
cloud	public	models.atlas.example/shared/cloud-platform@v0	v0.1.0	atlas-shared-cloud-platform

Alias	Publication	Module	Version	Source model
compliance	public	models.atlas.example/shared/compliance-baseline@v0	v0.1.0	atlas-shared-compliance-baseline
edge	public	models.atlas.example/shared/edge-platform@v0	v0.1.0	atlas-shared-edge-platform
security	public	models.atlas.example/shared/security-services@v0	v0.1.0	atlas-shared-security-services

8.4.1. Published Architecture

Qualified ID	Name	Publication	Source
cloud::ACT-CLOUD-OPERATOR	Cloud Operator	public	models.atlas.example/shared/cloud-platform@v0@v0.1.0
cloud::CAP-CLOUD-OPERATIONS	capability	public	models.atlas.example/shared/cloud-platform@v0@v0.1.0
cloud::DO-CLOUD-EVENT	Cloud Event	public	models.atlas.example/shared/cloud-platform@v0@v0.1.0
cloud::FG-CLOUD	Cloud Platform	public	models.atlas.example/shared/cloud-platform@v0@v0.1.0
cloud::FU-CLOUD-RUNTIME	Cloud Runtime	public	models.atlas.example/shared/cloud-platform@v0@v0.1.0
cloud::HW-CLOUD-REGION	Cloud Region	public	models.atlas.example/shared/cloud-platform@v0@v0.1.0
cloud::IF-CLOUD-EVENTS	Cloud Event Ingestion	public	models.atlas.example/shared/cloud-platform@v0@v0.1.0
cloud::NEED-CLOUD-IDENTITY	capability	public	models.atlas.example/shared/cloud-platform@v0@v0.1.0
compliance::ACT-COMP-OFFICER	Compliance Officer	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::ACT-EMPLOYEE	Employee	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::ACT-LINE-MANAGER	Line Manager	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0

Qualified ID	Name	Publication	Source
compliance::ACT-PEOPLE-OPERATIONS	People Operations Partner	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::CAP-CONTROL-BASELINE	capability	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::DO-COMP-EVIDENCE	Compliance Evidence Package	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::DO-COMP-PERSONNEL-RECORD	Personnel Lifecycle Record	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::DO-COMP-TRAINING-RECORD	Security Awareness Training Record	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::FG-COMP	Compliance Assurance	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::FU-COMP-EVIDENCE	Compliance Evidence Service	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::FU-COMP-PEOPLE	People Security and Awareness	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
edge::ACT-EDGE-OPERATOR	Edge Operator	public	models.atlas.example/shared/edge-platform@v0@v0.1.0
edge::CAP-EDGE-COMPUTE	capability	public	models.atlas.example/shared/edge-platform@v0@v0.1.0
edge::DO-EDGE-TELEMETRY	Edge Telemetry Record	public	models.atlas.example/shared/edge-platform@v0@v0.1.0
edge::FG-EDGE	Edge Platform	public	models.atlas.example/shared/edge-platform@v0@v0.1.0
edge::FU-EDGE-RUNTIME	Edge Runtime	public	models.atlas.example/shared/edge-platform@v0@v0.1.0
edge::HW-EDGE-COMPUTE	Rugged Edge Compute Unit	public	models.atlas.example/shared/edge-platform@v0@v0.1.0
edge::IF-EDGE-TELEMETRY	Edge Telemetry Interface	public	models.atlas.example/shared/edge-platform@v0@v0.1.0
edge::NEED-EDGE-IDENTITY	capability	public	models.atlas.example/shared/edge-platform@v0@v0.1.0

Qualified ID	Name	Publication	Source
security::ACT-SEC-ANALYST	Security Analyst	public	models.atlas.example/shared/security-services@v0@v0.1.0
security::CAP-DEVICE-IDENTITY	capability	public	models.atlas.example/shared/security-services@v0@v0.1.0
security::CAP-ZERO-TRUST	capability	public	models.atlas.example/shared/security-services@v0@v0.1.0
security::DO-SEC-AUDIT-EVENT	Security Audit Event	public	models.atlas.example/shared/security-services@v0@v0.1.0
security::FG-SEC	Security Services	public	models.atlas.example/shared/security-services@v0@v0.1.0
security::FU-SEC-IDENTITY	Identity Service	public	models.atlas.example/shared/security-services@v0@v0.1.0
security::FU-SEC-MONITORING	Security Monitoring	public	models.atlas.example/shared/security-services@v0@v0.1.0
security::IF-SEC-TOKEN	Identity Token Interface	public	models.atlas.example/shared/security-services@v0@v0.1.0

8.4.2. Published Requirements

Qualified ID	Name	Publication	Source
cloud::REQ-CLOUD-001	The atlas cloud platform shall process each cloud event using the cloud runtime.	public	models.atlas.example/shared/cloud-platform@v0@v0.1.0
cloud::REQ-CLOUD-002	The atlas cloud platform shall authenticate the cloud runtime using workload identity.	public	models.atlas.example/shared/cloud-platform@v0@v0.1.0
compliance::REQ-COMP-001	The atlas compliance baseline shall retain compliance evidence using the compliance evidence service.	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0

Qualified ID	Name	Publication	Source
compliance::REQ-COMP-002	The atlas compliance baseline shall report control implementation status to the compliance officer.	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::REQ-COMP-003	The people security and awareness shall record employee screening status in the personnel lifecycle record before privileged access.	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::REQ-COMP-004	The people security and awareness shall record policy acknowledgment for every employee at onboarding.	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::REQ-COMP-005	The people security and awareness shall record annual employee completion in the security awareness training record.	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::REQ-COMP-006	The people security and awareness shall request a line manager access review following an employee role change.	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::REQ-COMP-007	The people security and awareness shall revoke employee access following employment termination.	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::REQ-COMP-008	The people security and awareness shall request people operations partner confirmation of company asset return following employment termination.	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0

Qualified ID	Name	Publication	Source
edge::REQ-EDGE-001	The atlas edge platform shall execute signed workloads on the edge runtime.	public	models.atlas.example/shared/edge-platform@v0@v0.1.0
edge::REQ-EDGE-002	The atlas edge platform shall authenticate the edge runtime using device identity.	public	models.atlas.example/shared/edge-platform@v0@v0.1.0
security::REQ-SEC-001	The atlas security services shall issue short-lived identity credentials using the identity service.	public	models.atlas.example/shared/security-services@v0@v0.1.0
security::REQ-SEC-002	When credential replay is detected, the atlas security services shall notify the security analyst.	public	models.atlas.example/shared/security-services@v0@v0.1.0

8.4.3. Published Assurance

Qualified ID	Name	Publication	Source
cloud::CTRL-CLOUD-BACKUP	Encrypted Backup	public	models.atlas.example/shared/cloud-platform@v0@v0.1.0
compliance::CTRL-COMP-ACCESS-REVIEW	Quarterly Access Review	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::CTRL-COMP-EVIDENCE-RETENTION	Evidence Retention	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::CTRL-COMP-OFFBOARDING	Personnel Offboarding	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::CTRL-COMP-PERSONNEL-SCREENING	Personnel Screening	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::CTRL-COMP-POLICY-ACKNOWLEDGMENT	Employment Security Responsibilities	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::CTRL-COMP-ROLE-CHANGE-REVIEW	Role Change Review	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0

Qualified ID	Name	Publication	Source
compliance::CTRL-COMP-SECURITY-AWARENESS	Security Awareness Education	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::CV-COMP-ACCESS-REVIEW	-	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::CV-COMP-AWARENESS	-	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::CV-COMP-OFFBOARDING	-	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::CV-COMP-PERSONNEL-SCREENING	-	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::CV-COMP-ROLE-CHANGE	-	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::POAM-COMP-001	-	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::POAM-COMP-002	-	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::RISK-COMP-ACCESS-REVIEW-GAP	Delayed privileged access review	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::RISK-COMP-RETAINED-ACCESS	Former or transferred personnel retain access	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::RISK-COMP-UNTRAINED-PERSONNEL	Personnel miss security obligations	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
edge::CTRL-EDGE-SIGNED-WORKLOAD	Signed Workload Enforcement	public	models.atlas.example/shared/edge-platform@v0@v0.1.0
security::AV-CREDENTIAL-REPLAY	Credential Replay	public	models.atlas.example/shared/security-services@v0@v0.1.0
security::CTRL-SEC-AUDIT-INTEGRITY	Audit Integrity	public	models.atlas.example/shared/security-services@v0@v0.1.0
security::CTRL-SEC-SHORT-LIVED-IDENTITY	Short-lived Identity	public	models.atlas.example/shared/security-services@v0@v0.1.0
security::CV-SEC-TOKEN-REPLAY	-	public	models.atlas.example/shared/security-services@v0@v0.1.0

Qualified ID	Name	Publication	Source
security::MIT-SEC-TOKEN-REPLAY	-	public	models.atlas.example/shared/security-services@v0@v0.1.0
security::RISK-SEC-CREDENTIAL-REPLAY	Credential replay risk	public	models.atlas.example/shared/security-services@v0@v0.1.0
security::TB-SEC-CONTROL-PLANE	Security Control Plane	public	models.atlas.example/shared/security-services@v0@v0.1.0
security::TS-SEC-CREDENTIAL-REPLAY	Captured token is replayed	public	models.atlas.example/shared/security-services@v0@v0.1.0

8.4.4. Published Compliance

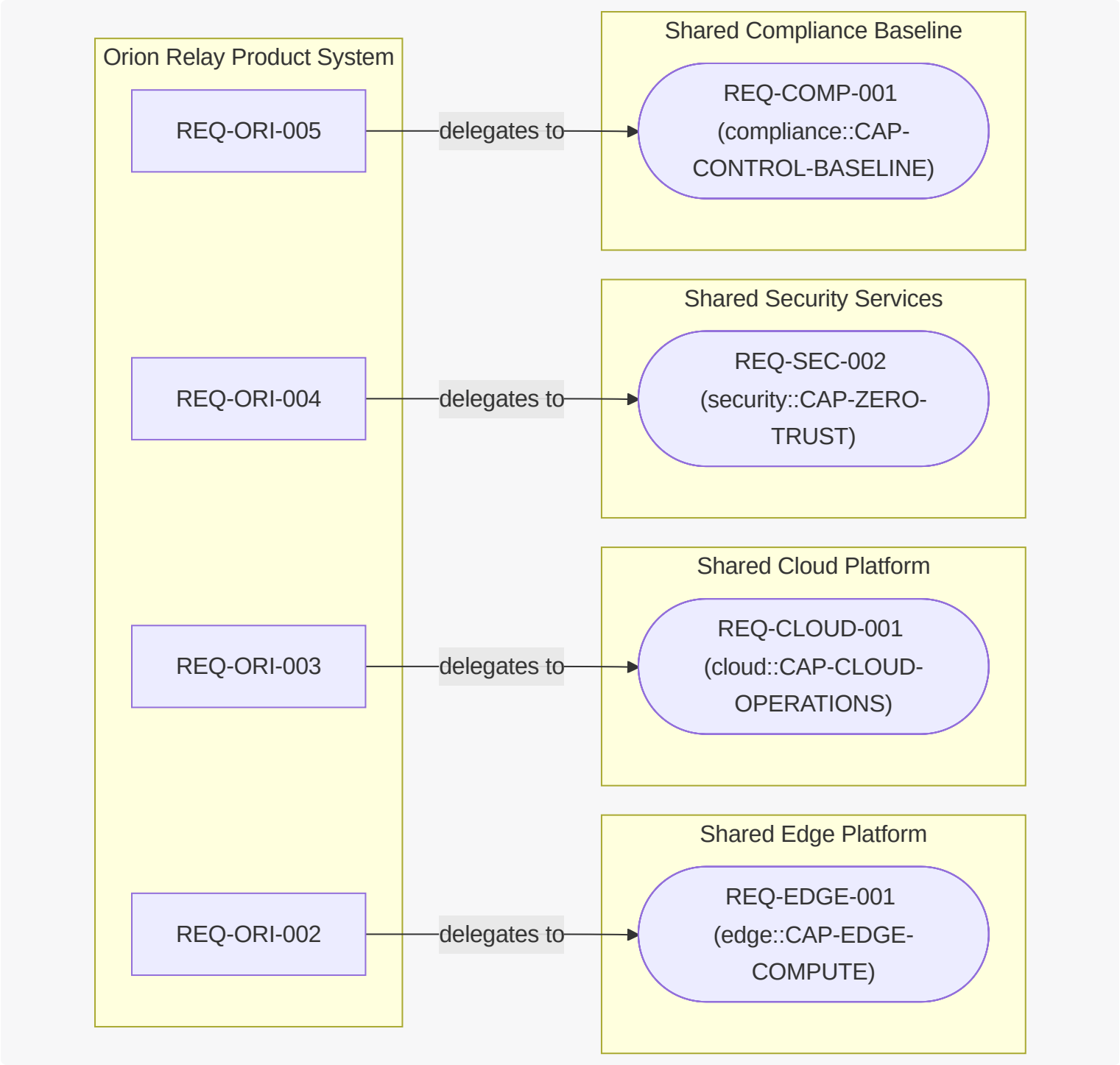
Qualified ID	Name	Publication	Source
compliance::IMPL-COMP-A5	-	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::IMPL-COMP-AC2	-	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::IMPL-COMP-ISO-PEOPLE	-	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::IMPL-COMP-NIST-PEOPLE	-	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::PROF-ISO-ATLAS	-	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0
compliance::PROF-NIST-ATLAS	-	public	models.atlas.example/shared/compliance-baseline@v0@v0.1.0

8.5. Requirement Allocation Matrix

Requirement	Subsystem	Target	Resolved
REQ-ORI-002	SUB-ORI-EDGE	edge::CAP-EDGE-COMPUTE	yes
REQ-ORI-003	SUB-ORI-CLOUD	cloud::CAP-CLOUD-OPERATIONS	yes
REQ-ORI-004	SUB-ORI-SECURITY	security::CAP-ZERO-TRUST	yes
REQ-ORI-005	SUB-ORI-COMPLIANCE	compliance::CAP-CONTROL-BASELINE	yes

8.5.1. Requirement Delegation View

Each delegated requirement of this system links downward to the subsystem contract it binds. Requirements not shown here carry no delegation link and are implemented in this system.



All subsystem required interfaces are satisfied and all allocations resolve to published contract identifiers.

9. Reference Index

9.1. Authored References

CTRL-ORI-ADMISSION

category=availability; description=Reserves capacity for authenticated priority messages.

Key	Value
Kind	Control
Name	Priority Admission Control
Mentioned In	n/a

DO-ORI-MESSAGE

termRef=DATA-ORI-MESSAGE; schemaRef=n/a; sensitivity=restricted

Key	Value
Kind	Data Object
Name	Relay Message
Mentioned In	REQ-ORI-001

IF-ORI-MESSAGE

protocol=quic; endpoint=quic://relay/messages; schemaRef=n/a; owner=FU-ORI-ROUTER

Key	Value
Kind	Interface
Name	Relay Message Interface
Mentioned In	n/a

REQ-ORI-001

The orion relay system shall deliver each relay message according to network controller priority.

Key	Value
Kind	Requirement
Name	REQ-ORI-001

Key	Value
Mentioned In	n/a

REQ-ORI-002

The orion relay system shall execute the link scheduler software on managed edge compute.

Key	Value
Kind	Requirement
Name	REQ-ORI-002
Mentioned In	n/a

REQ-ORI-003

The orion relay system shall retain relay configuration using managed cloud operations.

Key	Value
Kind	Requirement
Name	REQ-ORI-003
Mentioned In	n/a

REQ-ORI-004

The orion relay system shall authenticate product services using shared security services.

Key	Value
Kind	Requirement
Name	REQ-ORI-004
Mentioned In	n/a

REQ-ORI-005

The orion relay system shall retain compliance evidence using the company control baseline.

Key	Value
Kind	Requirement
Name	REQ-ORI-005
Mentioned In	n/a

9.2. Catalog References

ACT-ORI-CONTROLLER

Configures relay priorities and monitors delivery.

Key	Value
Kind	Catalog Actor
Name	network controller
Mentioned In	REQ-ORI-001

ACT-ORI-ENGINEER

Owns Orion security controls and risk.

Key	Value
Kind	Catalog Actor
Name	relay security engineer
Mentioned In	n/a

AV-ORI-MESSAGE-FLOOD

Excess messages attempt to exhaust constrained link capacity.

Key	Value
Kind	Catalog Attack Vector
Name	message flooding
Mentioned In	n/a

DATA-ORI-MESSAGE

Prioritized payload awaiting delivery across a remote link.

Key	Value
Kind	Catalog Data Term
Name	relay message
Mentioned In	n/a

FG-ORI

Product-specific link scheduling and message delivery.

Key	Value
Kind	Catalog Functional Group
Name	relay mission
Mentioned In	n/a

FU-ORI-LINK

Selects available radio links and transmission windows.

Key	Value
Kind	Catalog Functional Unit
Name	link scheduler software
Mentioned In	REQ-ORI-002

FU-ORI-ROUTER

Stores, prioritizes, and forwards operational messages.

Key	Value
Kind	Catalog Functional Unit
Name	message router software
Mentioned In	n/a

SYS-ORI

Resilient remote communications and store-and-forward product.

Key	Value
Kind	Catalog System
Name	orion relay system
Mentioned In	REQ-ORI-001 , REQ-ORI-002 , REQ-ORI-003 , REQ-ORI-004 , REQ-ORI-005

TERM-ACTOR

A person or operational role that interacts with functional units.

Key	Value
Kind	Engineering Model Term
Name	actor
Mentioned In	n/a

TERM-ASCIIDOC-DIAGRAM

Generated diagram block, usually Mermaid, that visualizes architecture relationships in the AsciiDoc publication.

Key	Value
Kind	Engineering Model Term
Name	AsciiDoc diagram
Mentioned In	n/a

TERM-ASCIIDOC-DOCUMENT

Human-readable generated architecture publication document assembled from authored model, inferred evidence, diagrams, references, and decisions.

Key	Value
Kind	Engineering Model Term
Name	AsciiDoc document
Mentioned In	n/a

TERM-ASCIIDOC-SECTION

Structured generated document section or row model used to render authored and inferred architecture content.

Key	Value
Kind	Engineering Model Term
Name	AsciiDoc section
Mentioned In	n/a

TERM-ATTACK-VECTOR

A technical misuse or attack path that targets functional, referenced, or runtime elements.

Key	Value
Kind	Engineering Model Term
Name	attack vector
Mentioned In	n/a

TERM-AUTHORED-MAPPING

An explicit relationship declared in architecture inputs between authored or referenced elements.

Key	Value
Kind	Engineering Model Term
Name	authored mapping
Mentioned In	n/a

TERM-BUNDLE

Loaded architecture, catalog, and companion documents resolved as one working model context.

Key	Value
Kind	Engineering Model Term
Name	model bundle
Mentioned In	n/a

TERM-CATALOG

Controlled vocabulary document used to normalize model, requirement, and generated-document terminology.

Key	Value
Kind	Engineering Model Term
Name	catalog
Mentioned In	n/a

TERM-CATALOG-ENTRY

A stable catalog term with a definition and aliases for linting, linking, and generated references.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	catalog entry
Mentioned In	n/a

TERM-CLI-COMMAND

Command-line entrypoint that coordinates loading, validation, generation, and file output for an engineering-model workflow.

Key	Value
Kind	Engineering Model Term
Name	CLI command
Mentioned In	n/a

TERM-CODE-ELEMENT

An inferred code structure or ownership element discovered from source trees and build metadata.

Key	Value
Kind	Engineering Model Term
Name	code element
Mentioned In	n/a

TERM-COMPLIANCE-MAPPING

Authored mapping that links a model control to selected compliance controls, model scope, implementation status, evidence, and responsible roles.

Key	Value
Kind	Engineering Model Term
Name	compliance mapping
Mentioned In	n/a

TERM-CONTROL

An authored security, policy, or operational control used to constrain behavior and reduce risk.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	control
Mentioned In	REQ-ORI-005

TERM-CONTROL-VERIFICATION

Authored control verification record with method, status, threat/risk scope, findings, and evidence.

Key	Value
Kind	Engineering Model Term
Name	control verification
Mentioned In	n/a

TERM-DATA-OBJECT

An authored data artifact or contract shape traced across flow, interface, and implementation boundaries.

Key	Value
Kind	Engineering Model Term
Name	data object
Mentioned In	n/a

TERM-DECISION

Authored architecture decision record with status, context, decision text, and consequences.

Key	Value
Kind	Engineering Model Term
Name	architecture decision
Mentioned In	n/a

TERM-DEPLOYMENT-TARGET

An authored deployment destination such as environment, cluster, namespace, or registry scope.

Key	Value
Kind	Engineering Model Term
Name	deployment target

Key	Value
Mentioned In	n/a

TERM-DESIGN

Authored design narrative organized by model entities and architecture views.

Key	Value
Kind	Engineering Model Term
Name	design document
Mentioned In	n/a

TERM-DESIGN-VIEW

View-scoped design narrative attached to an authored functional group or functional unit.

Key	Value
Kind	Engineering Model Term
Name	design view
Mentioned In	n/a

TERM-EVENT

An authored trigger signal that drives transitions, flow progress, or asynchronous outcomes.

Key	Value
Kind	Engineering Model Term
Name	event
Mentioned In	n/a

TERM-EVIDENCE

A file, artifact, or observation used to support control, risk, threat, or verification claims.

Key	Value
Kind	Engineering Model Term
Name	evidence
Mentioned In	REQ-ORI-005

TERM-FLOW

An authored causal interaction sequence from user/system intent to outcome, represented as ordered steps.

Key	Value
Kind	Engineering Model Term
Name	flow
Mentioned In	n/a

TERM-FLOW-STEP

A single authored step in a flow that captures action, data in/out, references, and normal/error/async transitions.

Key	Value
Kind	Engineering Model Term
Name	flow step
Mentioned In	n/a

TERM-FUNCTIONAL-GROUP

A major authored capability area that groups related functional units.

Key	Value
Kind	Engineering Model Term
Name	functional group
Mentioned In	n/a

TERM-FUNCTIONAL-UNIT

An authored working unit inside a functional group that owns specific behavior.

Key	Value
Kind	Engineering Model Term
Name	functional unit
Mentioned In	n/a

TERM-GENERATION-WORKFLOW

Orchestrated run that combines model loading, validation, projection, rendering, and artifact emission.

Key	Value
Kind	Engineering Model Term
Name	generation workflow
Mentioned In	n/a

TERM-INFERENCE-HINT

Authored source and ownership configuration used to discover runtime, code, and verification evidence.

Key	Value
Kind	Engineering Model Term
Name	inference hint
Mentioned In	n/a

TERM-INFERRED-MAPPING

A discovered relationship that links inferred runtime/code elements upward to authored design.

Key	Value
Kind	Engineering Model Term
Name	inferred mapping
Mentioned In	n/a

TERM-INTERFACE

An authored interface boundary (for example API, channel, endpoint, or contract) owned by a functional unit.

Key	Value
Kind	Engineering Model Term
Name	interface
Mentioned In	n/a

TERM-LINT-RUN

Requirement lint configuration that controls parsing and quality checks for requirement text.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	lint run
Mentioned In	n/a

TERM-LOBSTER-ACTIVITY-TRACE

Generated LOBSTER activity JSON that links verification evidence back to formal requirement identifiers.

Key	Value
Kind	Engineering Model Term
Name	LOBSTER activity trace
Mentioned In	n/a

TERM-MCP-SERVER

Model Context Protocol server that exposes engineering-model operations to AI agents through JSON-RPC tools.

Key	Value
Kind	Engineering Model Term
Name	MCP server
Mentioned In	n/a

TERM-MCP-STDIO-TRANSPORT

Content-Length framed standard input/output transport used by the MCP command-line server.

Key	Value
Kind	Engineering Model Term
Name	MCP stdio transport
Mentioned In	n/a

TERM-MCP-TOOL

Named MCP operation with a constrained input schema and structured response payload.

Key	Value
Kind	Engineering Model Term
Name	MCP tool

Key	Value
Mentioned In	n/a

TERM-MCP-TOOL-RESPONSE

Structured MCP tool payload that includes success state, schema version, generated timestamp, and result data or validation error.

Key	Value
Kind	Engineering Model Term
Name	MCP tool response
Mentioned In	n/a

TERM-MODEL

The authored architecture model root that composes functional structure, relationships, inference hints, and views.

Key	Value
Kind	Engineering Model Term
Name	model
Mentioned In	n/a

TERM-OPEN-OTM-DOCUMENT

Open Threat Model JSON representation generated from the engineering model for interoperable threat-model exchange.

Key	Value
Kind	Engineering Model Term
Name	Open OTM document
Mentioned In	n/a

TERM-OSCAL-ASSESSMENT-RESULTS

Generated OSCAL assessment results that summarize reviewed controls, verification findings, and modeled risks.

Key	Value
Kind	Engineering Model Term
Name	OSCAL assessment results

Key	Value
Mentioned In	n/a

TERM-OSCAL-POAM

Generated OSCAL plan of action and milestones that maps modeled POA&M items and related risks into compliance JSON.

Key	Value
Kind	Engineering Model Term
Name	OSCAL POA&M
Mentioned In	n/a

TERM-OSCAL-SSP

Generated OSCAL system security plan that maps authored system metadata, components, controls, allocations, and evidence into SSP JSON.

Key	Value
Kind	Engineering Model Term
Name	OSCAL SSP
Mentioned In	n/a

TERM-POAM-ITEM

Plan of action and milestones item linked to a modeled risk and supporting artifacts.

Key	Value
Kind	Engineering Model Term
Name	POA&M item
Mentioned In	n/a

TERM-REFERENCE-INDEX

Generated index of authored, catalog, runtime, code, and verification references with stable anchors and backlinks.

Key	Value
Kind	Engineering Model Term
Name	reference index

Key	Value
Mentioned In	n/a

TERM-REFERENCED-ELEMENT

An architecture-relevant external, platform, or third-party dependency represented by role.

Key	Value
Kind	Engineering Model Term
Name	referenced element
Mentioned In	n/a

TERM-REPO-INDEX

Bounded first-party source index used to answer model-aware file, requirement, control, and threat lookup queries.

Key	Value
Kind	Engineering Model Term
Name	repository index
Mentioned In	n/a

TERM-REQUIREMENT

A structured requirement statement that defines expected system behavior and maps to functional ownership.

Key	Value
Kind	Engineering Model Term
Name	requirement
Mentioned In	n/a

TERM-RISK

Authored risk record with likelihood, impact, response, scope, controls, and evidence.

Key	Value
Kind	Engineering Model Term
Name	risk
Mentioned In	n/a

TERM-RUNTIME-ELEMENT

An inferred runtime realization element discovered from infrastructure and deployment sources.

Key	Value
Kind	Engineering Model Term
Name	runtime element
Mentioned In	n/a

TERM-SECURITY-CONTEXT

Security-focused generated context view that groups owned functional units and shows external actors, references, flows, controls, and trust boundaries.

Key	Value
Kind	Engineering Model Term
Name	security context
Mentioned In	n/a

TERM-SOURCE-BLOCK

Stable source reference block that records file path, optional line span, kind, summary, and linked entity IDs.

Key	Value
Kind	Engineering Model Term
Name	source block
Mentioned In	n/a

TERM-STATE

An authored lifecycle state used to model transition behavior, guards, and event-driven progression.

Key	Value
Kind	Engineering Model Term
Name	state
Mentioned In	n/a

TERM-STRUCTURIZR-DSL

Generated Structurizr DSL workspace that projects authored architecture, relationships, dynamic views, and deployment metadata.

Key	Value
Kind	Engineering Model Term
Name	Structurizr DSL
Mentioned In	n/a

TERM-THREAT-ASSUMPTION

Authored security assumption that records scope, rationale, owner, status, and evidence.

Key	Value
Kind	Engineering Model Term
Name	threat assumption
Mentioned In	n/a

TERM-THREAT-DRAGON-DOCUMENT

Threat Dragon JSON representation generated from the engineering model for STRIDE threat-model review.

Key	Value
Kind	Engineering Model Term
Name	Threat Dragon document
Mentioned In	n/a

TERM-THREAT-MITIGATION

Authored mitigation record that links a threat scenario to a control, effectiveness, verification, and evidence.

Key	Value
Kind	Engineering Model Term
Name	threat mitigation
Mentioned In	n/a

TERM-THREAT-MODEL-EXPORT

Generated security artifact that translates authored architecture, flows, trust boundaries, threats, and mitigations into an external threat-model schema.

Key	Value
Kind	Engineering Model Term
Name	threat model export
Mentioned In	n/a

TERM-THREAT-OUT-OF-SCOPE

Authored decision that excludes a threat concern from current scope with reason, owner, expiry, and evidence.

Key	Value
Kind	Engineering Model Term
Name	threat out-of-scope decision
Mentioned In	n/a

TERM-THREAT-SCENARIO

Authored threat narrative that connects attack vectors, scope, flows, controls, risks, and verification evidence.

Key	Value
Kind	Engineering Model Term
Name	threat scenario
Mentioned In	n/a

TERM-TRACE-MARKER

Source-level marker such as TRLC-LINKS or ENGMODEL-LINKS used to connect declarations to requirements and model entities.

Key	Value
Kind	Engineering Model Term
Name	trace marker
Mentioned In	n/a

TERM-TRLC-PACKAGE

Generated TRLC model and requirements package used for formal requirement traceability processing.

Key	Value
Kind	Engineering Model Term
Name	TRLC package
Mentioned In	n/a

TERM-TRUST-BOUNDARY

An authored trust separation zone that marks policy, identity, or security control boundaries.

Key	Value
Kind	Engineering Model Term
Name	trust boundary
Mentioned In	n/a

TERM-UPWARD-LINKING

Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.

Key	Value
Kind	Engineering Model Term
Name	upward linking
Mentioned In	n/a

TERM-VALIDATION

Model quality gate that checks authored documents, references, relationship semantics, and requirement lint results.

Key	Value
Kind	Engineering Model Term
Name	validation
Mentioned In	n/a

TERM-VALIDATION-DIAGNOSTIC

Structured validation finding with code, severity, message, and optional source path.

Key	Value
Kind	Engineering Model Term
Name	validation diagnostic
Mentioned In	n/a

TERM-VERIFICATION-CHECK

An inferred or authored verification artifact that validates requirement behavior with test evidence.

Key	Value
Kind	Engineering Model Term
Name	verification check
Mentioned In	n/a

TERM-VIEW

Authored projection configuration that selects roots, entity kinds, mappings, audience, and abstraction.

Key	Value
Kind	Engineering Model Term
Name	view
Mentioned In	n/a

9.3. Inferred Runtime References

9.4. Inferred Code References

9.5. Verification References